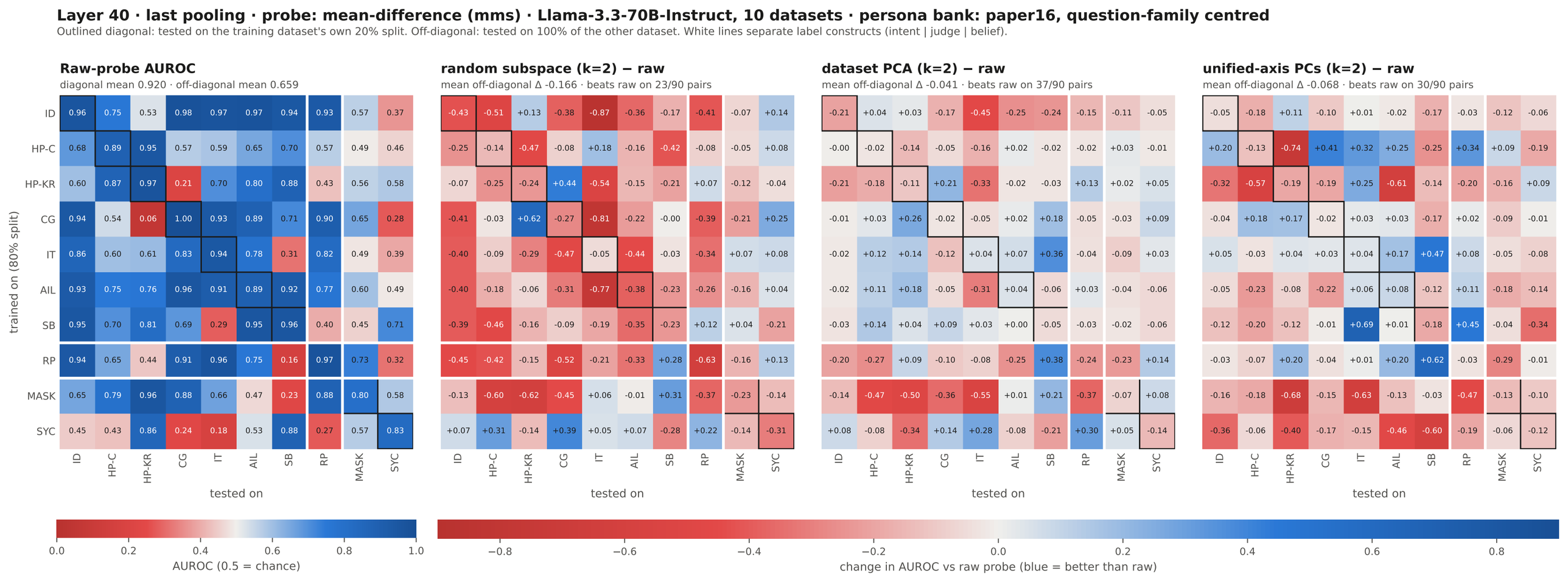

Layer 40 · last pooling · probe: mean-difference (mms) · Llama-3.3-70B-Instruct, 10 datasets · persona bank: paper16, question-family centred
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.920 · off-diagonal mean 0.659
random subspace (k=2) − raw
mean off-diagonal Δ -0.166 · beats raw on 23/90 pairs
dataset PCA (k=2) − raw
mean off-diagonal Δ -0.041 · beats raw on 37/90 pairs
unified-axis PCs (k=2) − raw
mean off-diagonal Δ -0.068 · beats raw on 30/90 pairs
trained on (80% split)
ID HP-C HP-KR CG IT AIL SB RP MASK SYC
tested on
AUROC (0.5 = chance)
change in AUROC vs raw probe (blue = better than raw)